

Instagram User Analytics

Project Description

This project focuses on analyzing Instagram user data using SQL queries in MySQL Workbench. The objective is to derive meaningful business insights related to user engagement, marketing strategies, investor metrics, and fake account detection.

The analysis helps different teams:

- Marketing Team → User engagement campaigns
- Product Team → Feature planning
- Investors → Platform activity and growth analysis

Tech-Stack Used

Tools	Purpose
MySQL Workbench	Database querying
SQL	Data analysis
PowerPoint / PDF	Reporting
Instagram Database	Dataset

(A) Marketing Analysis:

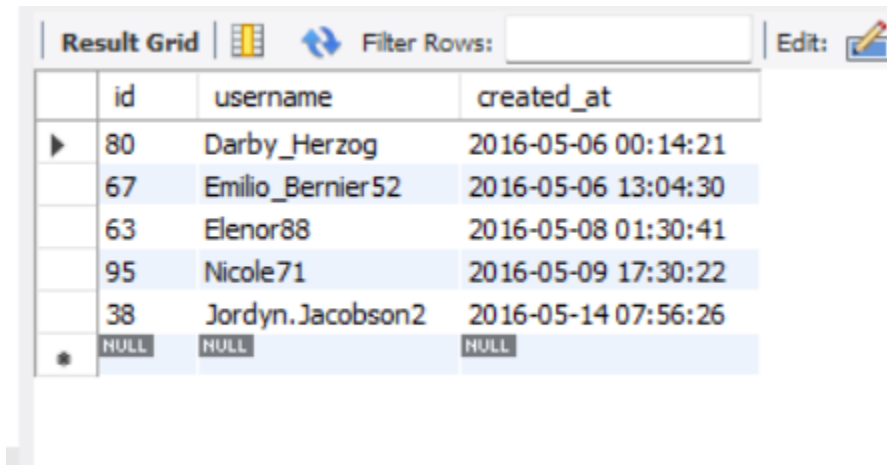
Task 1: Loyal User Reward

Goal: Identify the 5 oldest users.

SQL Query:

```
SELECT id, username, created_at
```

```
FROM users
ORDER BY created_at ASC
LIMIT 5;
```



The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' search bar and an 'Edit' button with a pencil icon. The table has four columns: an index column, 'id', 'username', and 'created_at'. The first five rows are highlighted in blue. The last row shows 'NULL' values for 'id', 'username', and 'created_at'.

	id	username	created_at
▶	80	Darby_Herzog	2016-05-06 00:14:21
	67	Emilio_Bernier52	2016-05-06 13:04:30
	63	Elenor88	2016-05-08 01:30:41
	95	Nicole71	2016-05-09 17:30:22
	38	Jordyn.Jacobson2	2016-05-14 07:56:26
✱	NULL	NULL	NULL

Task 2: Inactive User Engagement

Goal: Identify users who have never posted a photo.

SQL Query:

```
SELECT users.id, users.username
FROM users
LEFT JOIN photos
ON users.id = photos.user_id
WHERE photos.id IS NULL;
```

Result Grid			Filter Rows:
	id	username	
▶	5	Aniya_Hackett	
	7	Kassandra_Homenick	
	14	Jadyn81	
	21	Rocio33	
	24	Maxwell.Halvorson	
	25	Tierra.Trantow	
	34	Pearl7	
	36	Ollie_Ledner37	
	41	Mckenna17	
	45	David.Osinski47	
	49	Morgan.Kassulke	
	53	Linnea59	
	54	Duane60	
	57	Julien_Schmidt	
	66	Mike_Auer30	

Task 3: Contest Winner Declaration

Goal: Find the user with the most likes on a single photo.

SQL Query:

```
SELECT users.id, users.username, photos.id AS photo_id, photos.image_url,
       COUNT(likes.user_id) AS total_likes FROM photos
JOIN likes ON photos.id = likes.photo_id
JOIN users ON photos.user_id = users.id
GROUP BY photos.id
ORDER BY total_likes DESC
LIMIT 1;
```

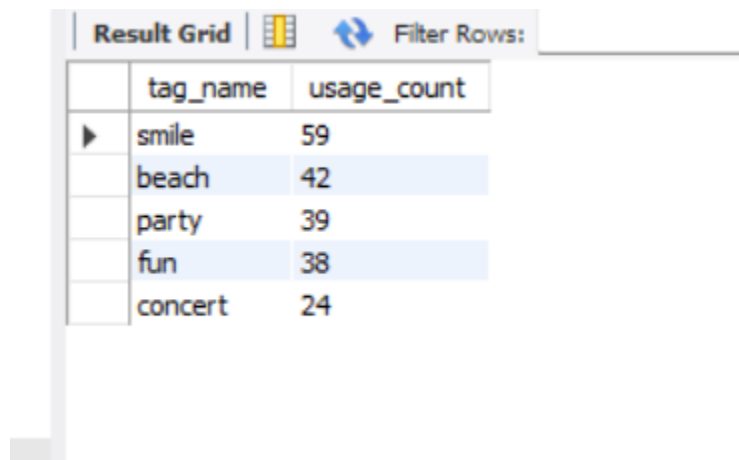
Result Grid						Filter Rows:	Export:	Wrap Cell Content:
	id	username	photo_id	image_url	total_likes			
▶	52	Zack_Kemmer93	145	https://jarret.name	48			

Task 4: Hashtag Research

Goal: Top 5 most commonly used hashtags.

SQL Query:

```
SELECT tags.tag_name,  
       COUNT(photo_tags.tag_id) AS usage_count  
FROM tags  
JOIN photo_tags ON tags.id = photo_tags.tag_id  
GROUP BY tags.id  
ORDER BY usage_count DESC  
LIMIT 5;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of the SQL query, showing the top 5 most commonly used hashtags. The columns are 'tag_name' and 'usage_count'. The rows are: 'smile' (59), 'beach' (42), 'party' (39), 'fun' (38), and 'concert' (24). The first row is highlighted with a blue background.

	tag_name	usage_count
▶	smile	59
	beach	42
	party	39
	fun	38
	concert	24

Task 5: Ad Campaign Launch

Goal: Best day of the week for registrations.

SQL Query:

```
SELECT DAYNAME(created_at) AS registration_day,  
       COUNT(*) AS total_users  
FROM users  
GROUP BY registration_day  
ORDER BY total_users DESC;
```

Result Grid			Filter Rows:
	registration_day	total_users	
▶	Thursday	16	
	Sunday	16	
	Friday	15	
	Tuesday	14	
	Monday	14	
	Wednesday	13	
	Saturday	12	

(B) Investor Metrics

Task 1: User Engagement

Goal: Average posts per user.

SQL Query:

```
SELECT
    (SELECT COUNT(*) FROM photos) / (SELECT COUNT(*) FROM users) AS
    avg_posts_per_user;
```

Result Grid			Filter Rows:
	avg_posts_per_user		
▶	2.5700		

Task 2: Bots & Fake Accounts

Goal: Identify users who liked every photo.

SQL Query:

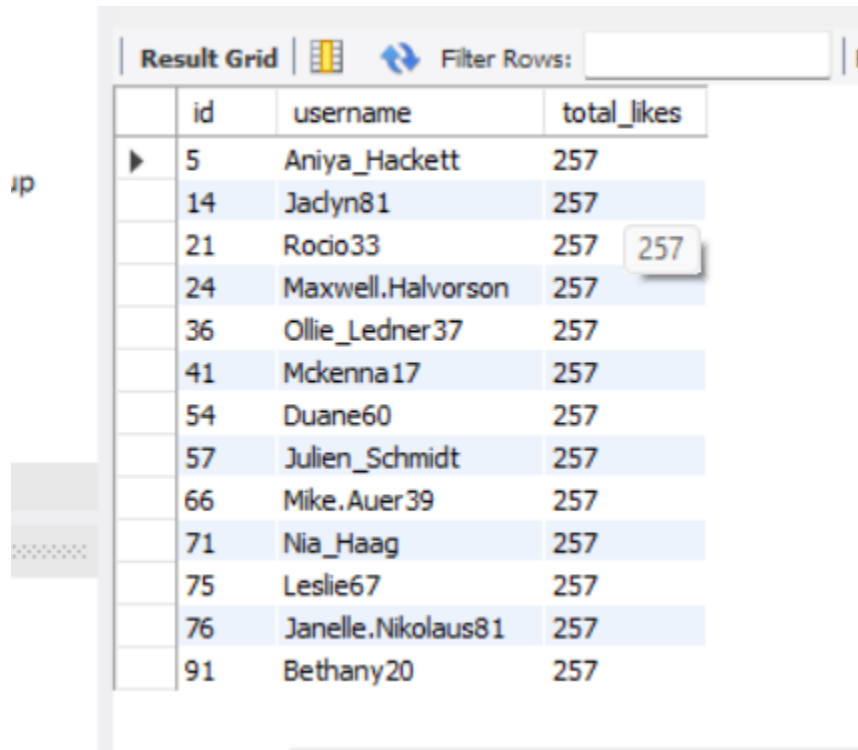
```
SELECT users.id,
    users.username,
    COUNT(likes.photo_id) AS total_likes
```

FROM users

JOIN likes ON users.id = likes.user_id

GROUP BY users.id

HAVING total_likes = (SELECT COUNT(*) FROM photos);



Result Grid | Filter Rows:

	id	username	total_likes
▶	5	Aniya_Hackett	257
	14	Jadyn81	257
	21	Rocio33	257
	24	Maxwell.Halvorson	257
	36	Ollie_Ledner37	257
	41	Mckenna17	257
	54	Duane60	257
	57	Julien_Schmidt	257
	66	Mike.Auer39	257
	71	Nia_Haag	257
	75	Leslie67	257
	76	Janelle.Nikolaus81	257
	91	Bethany20	257

Project Approach

Step-by-Step Approach

1. Created database using provided SQL file
2. Imported tables into MySQL Workbench
3. Understood table relationships
4. Used:
 - a) JOIN
 - b) GROUP BY
 - c) ORDER BY
 - d) Aggregate Functions
 - e) Subqueries
5. Extracted business insights from SQL outputs
6. Prepared report/PPT with screenshots

Key Insights

Marketing Insights

- Loyal users can be rewarded
- Inactive users can be targeted through campaigns
- Trending hashtags improve reach
- Best ad timing identified

Investor Insights

- Average posting rate measures engagement
- Fake accounts detected through abnormal behavior
- Popular content creators identified

Results

Achievements

- Performed real-world SQL analysis
- Improved understanding of:
 - Joins
 - Aggregations
 - Subqueries
 - Business analytics
- Derived actionable insights from user data

Impact

This project demonstrates how SQL can help businesses:

- Understand customer behavior
- Improve marketing strategy
- Detect suspicious accounts
- Increase engagement